Vocabulary = { cat, mat, the, chair, table, on, floor, sat }

Cat = [1, 0, 0, 0, 0, 0, 0, 0]
Mat = [0, 1, 0, 0, 0, 0, 0, 0]
the = [0, 0, 1, 0, 0, 0, 0, 0]
chair = [0, 0, 0, 1, 0, 0, 0, 0]
table = [0, 0, 0, 0, 1, 0, 0, 0]
on = [0, 0, 0, 0, 0, 1, 0, 0]
floor = [0, 0, 0, 0, 0, 0, 1, 0]
sat = [0, 0, 0, 0, 0, 0, 0, 1]

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} -0.22 & 0.71 & 0.47 & -0.07 \\ -0.85 & -1.51 & -0.45 & 0.86 \\ 0.21 & -1.25 & 0.17 & 0.39 \\ -0.88 & 0.15 & 0.06 & -1.14 \\ 0.36 & 0.56 & 1.08 & 1.05 \\ -1.38 & -0.94 & 0.52 & 0.51 \\ 0.52 & 3.85 & 0.57 & 1.14 \\ 0.95 & 0.65 & -0.32 & 0.76 \end{bmatrix}$$

$$\begin{array}{l} \text{Cat} \rightarrow \\ \text{Mat} \rightarrow \\ \text{the} \rightarrow \\ \text{chair} \rightarrow \\ \text{table} \rightarrow \\ \text{on} \rightarrow \\ \text{floor} \rightarrow \\ \text{sat} \rightarrow \end{array} \begin{bmatrix} -0.22 & 0.71 & 0.47 & -0.07 \\ -0.85 & -1.51 & -0.45 & 0.86 \\ 0.21 & -1.25 & 0.17 & 0.39 \\ -0.88 & 0.15 & 0.06 & -1.14 \\ 0.36 & 0.56 & 1.08 & 1.05 \\ -1.38 & -0.94 & 0.52 & 0.51 \\ 0.52 & 3.85 & 0.57 & 1.14 \\ 0.95 & 0.65 & -0.32 & 0.76 \end{bmatrix}$$

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LSTM forward pass with given words

$h_{t-1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Concatenate both vectors

the $= x_t = \begin{bmatrix} 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix}$

$$I = [h_{t-1}, x_t]$$

$$I = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix}$$

$$i_t = \text{Sigmoid}(W_i * I + B_i)$$

$$W_i * I = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -2.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix}$$

$$= \begin{bmatrix} -0.1945 \\ 2.2237 \\ 0.6494 \\ -1.20214 \end{bmatrix}$$

$$W_i * I + B = \begin{bmatrix} -0.1945 \\ 2.2237 \\ 0.6494 \\ -1.20214 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 0.1655 \\ 1.5737 \\ 1.0094 \\ 0.29786 \end{bmatrix}$$

$$\text{Sigmoid}(W_i * I + B) = \begin{bmatrix} 0.54 \\ 0.82 \\ 0.73 \\ 0.57 \end{bmatrix} = i_t$$

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$$f_t = \text{Sigmoid}(W_f * I + B_f)$$

$$W_f * I = \begin{bmatrix} 0.5 & -0.14 & 0.65 & 1.5 & -0.23 & -0.23 & 1.6 & 0.77 \\ -0.47 & 0.54 & -0.46 & -0.47 & 0.24 & -1.9 & -1.7 & -0.56 \\ -1 & 0.31 & -0.91 & -1.4 & 1.5 & -0.23 & 0.068 & -1.4 \\ -0.54 & 0.11 & -1.2 & 0.38 & -0.6 & -0.29 & -0.6 & 1.9 \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8115 \\ 1.918 \\ 0.06806 \\ 0.8755 \end{bmatrix}$$

$$W_f * I + B_f = \begin{bmatrix} 0.8115 \\ 1.918 \\ 0.06806 \\ 0.8755 \end{bmatrix} + \begin{bmatrix} -0.013 \\ -1.1 \\ 0.82 \\ -1.2 \end{bmatrix}$$

$$\begin{bmatrix} 0.7985 \\ 0.818 \\ 0.88806 \\ -0.3245 \end{bmatrix}$$

$$f_t = \text{Sigmoid}(W_f * I + B_f) = \begin{bmatrix} 0.68 \\ 0.69 \\ 0.70 \\ 0.58 \end{bmatrix}$$

$$\tilde{C}_t = \tanh(W_c * I + b_c)$$

$$W_c * I + b_c = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix}$$

$$= \begin{bmatrix} -0.1945 \\ 2.2237 \\ 0.6494 \\ -1.20214 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 0.1655 \\ 1.5737 \\ 1.0094 \\ 0.29786 \end{bmatrix}$$

$$\tanh(W_c * I + b_c) = \tilde{C}_t = \begin{bmatrix} 0.16 \\ 0.91 \\ 0.76 \\ 0.28 \end{bmatrix}$$

$$f_t \circledast C_{t-1} = \begin{bmatrix} 0.68 \\ 0.69 \\ 0.70 \\ 0.58 \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$i_t \circledast \tilde{C}_t = \begin{bmatrix} 0.54 \\ 0.82 \\ 0.73 \\ 0.57 \end{bmatrix} \circledast \begin{bmatrix} 0.16 \\ 0.91 \\ 0.76 \\ 0.28 \end{bmatrix}$$

$$= \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix}$$

$$C_{t-1} = f_t \circledast C_{t-1} \oplus i_t \circledast \tilde{C_t}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \oplus \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix} = \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + B_o)$$

$$W_o * I = \begin{bmatrix} 0.26 & -0.074 & -1.9 & -0.027 & 0.06 & 2.5 & -0.19 & 0.3 \\ -0.035 & -1.2 & 1.1 & 0.75 & 0.79 & -0.91 & 1.4 & -1.4 \\ 0.59 & 2.2 & -0.99 & -0.57 & 0.1 & -0.5 & -1.6 & 0.069 \\ -1.1 & 0.47 & -0.92 & 1.5 & -0.78 & -0.32 & 0.81 & -1.2 \end{bmatrix}$$

$$* \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.21 \\ -1.25 \\ 0.17 \\ 0.39 \end{bmatrix} = \begin{bmatrix} -3.0277 \\ 0.9954 \\ 0.40091 \\ -0.0941 \end{bmatrix}$$

$$W_o * I + b_o = \begin{bmatrix} -3.0277 \\ 0.9954 \\ 0.40091 \\ -0.0941 \end{bmatrix} + \begin{bmatrix} 0.23 \\ 1.3 \\ -1.6 \\ 0.18 \end{bmatrix}$$

$$= \begin{bmatrix} -2.79 \\ 2.29 \\ -1.19 \\ 0.08 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + b_o) = \begin{bmatrix} 0.05 \\ 0.90 \\ 0.23 \\ 0.51 \end{bmatrix}$$

$$C_t = \text{Tanh} \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix} = \begin{bmatrix} 0.07 \\ 0.62 \\ 0.50 \\ 0.14 \end{bmatrix}$$

$$h_t = O_t \circledast C_t$$

$$= \begin{bmatrix} 0.05 \\ 0.90 \\ 0.23 \\ 0.51 \end{bmatrix} \circledast \begin{bmatrix} 0.07 \\ 0.62 \\ 0.50 \\ 0.14 \end{bmatrix} = \begin{bmatrix} 0.0035 \\ 0.558 \\ 0.115 \\ 0.0714 \end{bmatrix}$$

$$h_{t-1} = \begin{bmatrix} 0.0035 \\ 0.558 \\ 0.115 \\ 0.0714 \end{bmatrix} \xrightarrow[\text{Precision}]{\text{Two Point}} \begin{bmatrix} 0.00 \\ 0.55 \\ 0.11 \\ 0.07 \end{bmatrix}$$

$$C_{t-1} = \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix}$$

$$X_t = \text{Cat} = \begin{bmatrix} -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix}$$

$$I = h_{t-1}, x_t$$

concatenate

$$I = \begin{bmatrix} 0.00 \\ 0.55 \\ 0.11 \\ 0.07 \\ -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix}$$

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$$it = \text{Sigmoid}(Wi * I + Bi)$$

$$Wi * I = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix}$$

$$* \begin{bmatrix} 0.00 \\ 0.55 \\ 0.11 \\ 0.07 \\ -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix} = \begin{bmatrix} -1.30 \\ -1.54 \\ 0.56 \\ 0.40 \end{bmatrix}$$

$$Wi * I + Bi = \begin{bmatrix} -1.30 \\ -1.54 \\ 0.56 \\ 0.40 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} -0.94 \\ -2.19 \\ 0.92 \\ 1.9 \end{bmatrix}$$

$$it = \text{Sigmoid}(Wi * I + Bi) = \begin{bmatrix} 0.28 \\ 0.10 \\ 0.71 \\ 0.86 \end{bmatrix}$$

$$ft = \text{Sigmoid}(Wf * I + Bf)$$

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$$W_f * I = \begin{bmatrix} 0.5 & -0.14 & 0.65 & 1.5 & -0.23 & -0.23 & 1.6 & 0.77 \\ -0.47 & 0.54 & -0.46 & -0.47 & 0.24 & -1.9 & -1.7 & -0.56 \\ -1 & 0.31 & -0.91 & -1.4 & 1.5 & -0.23 & 0.068 & -1.4 \\ -0.54 & 0.11 & -1.2 & 0.38 & -0.6 & -0.29 & -0.6 & 1.9 \end{bmatrix} * \begin{bmatrix} 0 \\ 0.55 \\ 0.11 \\ 0.07 \\ -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix}$$

$$\begin{bmatrix} 0.6849 \\ -1.9481 \\ -0.39094 \\ -0.5338 \end{bmatrix}$$

$$W_f * I + B_f = \begin{bmatrix} 0.6849 \\ -1.9481 \\ -0.39094 \\ -0.5338 \end{bmatrix} + \begin{bmatrix} -0.013 \\ -1.1 \\ 0.82 \\ -1.2 \end{bmatrix}$$

$$\begin{bmatrix} 0.6719 \\ -3.0481 \\ 0.42906 \\ -1.7338 \end{bmatrix}$$

$$f_t = \text{Sigmoid}(W_f * I + B_f) = \begin{bmatrix} 0.66 \\ 0.04 \\ 0.60 \\ 0.15 \end{bmatrix}$$

$$\tilde{C_t} = \tanh(W_c * I + b_c)$$

$$W_c * I + b_c = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 0.55 \\ 0.11 \\ 0.07 \\ -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix}$$

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$$\begin{bmatrix} -1.30 \\ -1.54 \\ 0.56 \\ 0.40 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} -0.94 \\ -2.19 \\ 0.92 \\ 1.9 \end{bmatrix}$$

$$\tilde{c_t} = \tanh(W_c * I + B_c) = \begin{bmatrix} -0.73 \\ -0.97 \\ 0.72 \\ 0.95 \end{bmatrix}$$

$$f_t \circledast c_{t-1} = \begin{bmatrix} 0.66 \\ 0.04 \\ 0.60 \\ 0.15 \end{bmatrix} \circledast \begin{bmatrix} 0.08 \\ 0.74 \\ 0.55 \\ 0.15 \end{bmatrix}$$

$$= \begin{bmatrix} 0.05 \\ 0.02 \\ 0.33 \\ 0.02 \end{bmatrix}$$

$$i_t \circledast \tilde{c_t} = \begin{bmatrix} 0.28 \\ 0.10 \\ 0.71 \\ 0.86 \end{bmatrix} \circledast \begin{bmatrix} -0.73 \\ -0.97 \\ 0.72 \\ 0.95 \end{bmatrix}$$

$$= \begin{bmatrix} -0.20 \\ -0.09 \\ 0.511 \\ 0.81 \end{bmatrix}$$

$$c_{t-1} = f_t \circledast c_{t-1} \oplus i_t \circledast \tilde{c_t}$$

$$\begin{bmatrix} 0.05 \\ 0.02 \\ 0.33 \\ 0.02 \end{bmatrix} + \begin{bmatrix} -0.20 \\ -0.09 \\ 0.51 \\ 0.81 \end{bmatrix} = \begin{bmatrix} -0.15 \\ -0.07 \\ 0.84 \\ 0.83 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + B_o)$$

$$W_o * I = \begin{bmatrix} 0.26 & -0.074 & -1.9 & -0.027 & 0.06 & 2.5 & -0.19 & 0.3 \\ -0.035 & -1.2 & 1.1 & 0.75 & 0.79 & -0.91 & 1.4 & -1.4 \\ 0.59 & 2.2 & -0.99 & -0.57 & 0.1 & -0.5 & -1.6 & 0.069 \\ -1.1 & 0.47 & -0.92 & 1.5 & -0.78 & -0.32 & 0.81 & -1.2 \end{bmatrix}$$

$$* \begin{bmatrix} 0 \\ 0.55 \\ 0.11 \\ 0.07 \\ -0.22 \\ 0.71 \\ 0.47 \\ -0.07 \end{bmatrix} = \begin{bmatrix} 1.39 \\ -0.55 \\ -0.07 \\ 0.67 \end{bmatrix}$$

$$W_o * I + b_o = \begin{bmatrix} 1.39 \\ -0.55 \\ -0.07 \\ 0.67 \end{bmatrix} + \begin{bmatrix} 0.23 \\ 1.3 \\ -1.6 \\ 0.18 \end{bmatrix}$$

$$= \begin{bmatrix} 1.62 \\ 0.75 \\ -1.67 \\ 0.85 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + b_o) = \begin{bmatrix} 0.83 \\ 0.67 \\ 0.15 \\ 0.70 \end{bmatrix}$$

$$c_{t-1} = \tanh\begin{pmatrix}-0.15\\-0.7\\0.84\\0.83\end{pmatrix} = \begin{bmatrix}-0.14\\-0.60\\0.68\\0.68\end{bmatrix}$$

$$h_t = O_t * c_t$$

$$\begin{bmatrix}0.83\\0.67\\0.15\\0.70\end{bmatrix} * \begin{bmatrix}-0.14\\-0.60\\0.68\\0.68\end{bmatrix} = \begin{bmatrix}-0.11\\-0.40\\0.10\\0.47\end{bmatrix}$$

$$h_{t-1} = \begin{bmatrix}-0.11\\-0.40\\0.10\\0.47\end{bmatrix}$$

$$c_{t-1} = \begin{bmatrix}-0.15\\-0.07\\0.84\\0.83\end{bmatrix}$$

$$\text{Sat} = x_t = \begin{bmatrix}0.95\\0.65\\-0.32\\0.76\end{bmatrix}$$

$$I = (\overbrace{h_{t-1}, x_t}^{\text{concatenate}}) = \begin{bmatrix}-0.11\\-0.40\\0.10\\0.47\\0.95\\0.65\\-0.32\\0.76\end{bmatrix}$$

$$i_t = \text{Sigmoid}(W_i * I + B_i)$$

$$W_i * I = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} * \begin{bmatrix} -0.11 \\ -0.40 \\ 0.10 \\ 0.47 \\ 0.95 \\ 0.65 \\ -0.32 \\ 0.76 \end{bmatrix}$$

$$\begin{bmatrix} 1.36 \\ -0.32 \\ 0.0076 \\ 1.91 \end{bmatrix}$$

$$(W_i * I + B_i) = \begin{bmatrix} 1.36 \\ -0.32 \\ 0.0076 \\ 1.91 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 1.72 \\ -0.97 \\ 0.36 \\ 3.41 \end{bmatrix}$$

$$\text{Sigmoid}(W_i * I + B_i) = \begin{bmatrix} 0.84 \\ 0.27 \\ 0.58 \\ 0.96 \end{bmatrix} = i_t$$

$$f_t = \text{Sigmoid}(W_f * I + B_f)$$

$$W_f * I = \begin{bmatrix} 0.5 & -0.14 & 0.65 & 1.5 & -0.23 & -0.23 & 1.6 & 0.77 \\ -0.47 & 0.54 & -0.46 & -0.47 & 0.24 & -1.9 & -1.7 & -0.56 \\ -1 & 0.31 & -0.91 & -1.4 & 1.5 & -0.23 & 0.068 & -1.4 \\ -0.54 & 0.11 & -1.2 & 0.38 & -0.6 & -0.29 & -0.6 & 1.9 \end{bmatrix} * \begin{bmatrix} -0.11 \\ -0.40 \\ 0.10 \\ 0.47 \\ 0.95 \\ 0.65 \\ -0.32 \\ 0.76 \end{bmatrix}$$

$$= \begin{bmatrix} 0.47 \\ -1.31 \\ -0.57 \\ 0.85 \end{bmatrix}$$

$$W_f * I + B_f = \begin{bmatrix} 0.47 \\ -1.31 \\ -0.57 \\ 0.95 \end{bmatrix} + \begin{bmatrix} -0.013 \\ -1.1 \\ 0.82 \\ -1.2 \end{bmatrix}$$

$$\begin{bmatrix} 0.45 \\ -2.41 \\ 0.25 \\ -0.25 \end{bmatrix}$$

$$f_t = \text{Sigmoid}(W_f * I + B_f) = \begin{bmatrix} 0.61 \\ 0.08 \\ 0.56 \\ 0.43 \end{bmatrix}$$

$$C'_t = \tanh(W_c * I + b_c)$$

$$W_c * I + b_c = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} * \begin{bmatrix} -0.11 \\ -0.40 \\ 0.10 \\ 0.47 \\ 0.95 \\ 0.65 \\ -0.32 \\ 0.76 \end{bmatrix}$$

$$W_c * I = \begin{bmatrix} 1.36 \\ -0.32 \\ 0.0076 \\ 1.91 \end{bmatrix}$$

$$W_c * I + b_c = \begin{bmatrix} 1.36 \\ -0.32 \\ 0.0076 \\ 1.91 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 1.72 \\ -0.97 \\ 0.36 \\ 3.41 \end{bmatrix}$$

$$\tanh(W_c * I + b_c) = \tilde{c_t} = \begin{bmatrix} 0.93 \\ -0.74 \\ 0.34 \\ 0.99 \end{bmatrix}$$

$$f_t \circledast c_{t-1} = \begin{bmatrix} 0.61 \\ 0.08 \\ 0.56 \\ 0.43 \end{bmatrix} \circledast \begin{bmatrix} -0.15 \\ -0.07 \\ 0.84 \\ 0.83 \end{bmatrix}$$

$$= \begin{bmatrix} -0.09 \\ 0.0056 \\ 0.47 \\ 0.35 \end{bmatrix}$$

$$i_t \circledast \tilde{c_t} = \begin{bmatrix} 0.84 \\ 0.27 \\ 0.58 \\ 0.96 \end{bmatrix} \circledast \begin{bmatrix} 0.93 \\ -0.74 \\ 0.34 \\ 0.99 \end{bmatrix}$$

$$= \begin{bmatrix} 0.78 \\ -0.19 \\ 0.19 \\ 0.95 \end{bmatrix}$$

$$c_{t-1} = f_t \circledast c_{t-1} \oplus i_t \circledast \tilde{c_t}$$

$$\begin{bmatrix} -0.09 \\ 0.0056 \\ 0.47 \\ 0.35 \end{bmatrix} \oplus \begin{bmatrix} 0.78 \\ -0.19 \\ 0.19 \\ 0.95 \end{bmatrix} = \begin{bmatrix} 0.69 \\ -0.1844 \\ 0.66 \\ 1.3 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + B_o)$$

$$W_o * I = \begin{bmatrix} 0.26 & -0.074 & -1.9 & -0.027 & 0.06 & 2.5 & -0.19 & 0.3 \\ -0.035 & -1.2 & 1.1 & 0.75 & 0.74 & -0.91 & 1.4 & -1.4 \\ 0.59 & 2.2 & -0.99 & -0.57 & 0.1 & -0.5 & -1.6 & 0.069 \\ -1.1 & 0.47 & -0.92 & 1.5 & -0.78 & -0.32 & 0.81 & -1.2 \end{bmatrix} * \begin{bmatrix} -0.17 \\ -0.40 \\ 0.10 \\ 0.47 \\ 0.95 \\ 0.65 \\ -0.32 \\ 0.76 \end{bmatrix}$$

$$\begin{bmatrix} 1.76 \\ -0.40 \\ -0.97 \\ -1.57 \end{bmatrix}$$

$$W_o * I + b_o = \begin{bmatrix} 1.76 \\ -0.40 \\ -0.97 \\ -1.57 \end{bmatrix} + \begin{bmatrix} 0.23 \\ 1.3 \\ -1.6 \\ 0.18 \end{bmatrix} = \begin{bmatrix} 1.99 \\ 0.9 \\ -2.57 \\ -1.39 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + b_o) = \begin{bmatrix} 0.87 \\ 0.71 \\ 0.07 \\ 0.19 \end{bmatrix}$$

$$c_{t-1} = \tanh\begin{pmatrix} 0.69 \\ -0.18 \\ 0.66 \\ 1.3 \end{pmatrix} = \begin{pmatrix} 0.59 \\ -0.17 \\ 0.57 \\ 0.86 \end{pmatrix}$$

$$h_t = O_t * C_t$$

$$\begin{bmatrix} 0.87 \\ 0.71 \\ 0.07 \\ 0.19 \end{bmatrix} * \begin{bmatrix} 0.59 \\ -0.17 \\ 0.57 \\ 0.86 \end{bmatrix} = \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \end{bmatrix}$$

$$h_{t-1} = \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \end{bmatrix}$$

$$c_{t-1} = \begin{bmatrix} 0.69 \\ -0.18 \\ 0.66 \\ 1.3 \end{bmatrix}$$

$$\text{on} = x_t = \begin{bmatrix} -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix}$$

$$I = [\, \underbrace{h_{t-1} \,,\, x_t}_{\text{concatenate}} \,]$$

$$I = \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \\ -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix}$$

$$i_t = \text{Sigmoid}\,(W_i * I + B_i)$$

$$W_i * I = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.93 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} * \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \\ -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix}$$

$$= \begin{bmatrix} -1.05 \\ 0.67 \\ 1.88 \\ -2.40 \end{bmatrix}$$

$$W_i * I + B_i = \begin{bmatrix} -1.05 \\ 0.67 \\ 1.88 \\ -2.40 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix}$$

$$= \begin{bmatrix} -0.69 \\ 0.02 \\ 2.24 \\ -0.74 \end{bmatrix}$$

$$i_t = \text{Sigmoid}(W_i * I + B_i) = \begin{bmatrix} 0.33 \\ 0.50 \\ 0.90 \\ 0.32 \end{bmatrix}$$

$$f_t = \text{Sigmoid}(W_f * I + B_f)$$

$$W_f * I = \begin{bmatrix} 0.5 & -0.14 & 0.65 & 1.5 & -0.23 & -0.23 & 1.6 & 0.77 \\ -0.47 & 0.54 & -0.46 & -0.47 & 0.24 & -1.9 & -1.7 & -0.56 \\ -1 & 0.31 & -0.91 & -1.4 & 1.5 & -0.23 & 0.068 & -1.4 \\ -0.54 & 0.11 & -1.2 & 0.38 & -0.6 & -0.29 & -0.6 & 1.9 \end{bmatrix}$$

$$* \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \\ -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix} = \begin{bmatrix} 2.28 \\ -0.10 \\ -3.33 \\ 1.49 \end{bmatrix}$$

$$Wf * I + Bf = \begin{bmatrix} 2.28 \\ -0.10 \\ -3.33 \\ 1.49 \end{bmatrix} + \begin{bmatrix} -0.013 \\ -1.1 \\ 0.82 \\ -1.2 \end{bmatrix}$$

$$= \begin{bmatrix} 2.267 \\ -1 \\ -2.51 \\ 0.29 \end{bmatrix}$$

$$f_t = \text{Sigmoid}(Wf * I + Bf) = \begin{bmatrix} 0.93 \\ 0.26 \\ 0.07 \\ 0.57 \end{bmatrix}$$

$$\tilde{C}t = \tanh(Wc * I + Bc)$$

$$Wc * I = \begin{bmatrix} 0.21 & -2 & -1.3 & 0.2 & 0.74 & 0.17 & -0.12 & -0.3 \\ -1.5 & -0.72 & -0.46 & 1.1 & 0.34 & -1.8 & 0.32 & -0.39 \\ -0.68 & 0.61 & 1 & 0.43 & -0.84 & -0.31 & 0.33 & 0.98 \\ -0.48 & -0.19 & -1.1 & -1.2 & 0.81 & 1.4 & -0.072 & 1 \end{bmatrix} *$$

$$\begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \\ -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix} = \begin{bmatrix} -1.05 \\ 0.67 \\ 1.88 \\ -2.40 \end{bmatrix}$$

$$Wc * I + Bc = \begin{bmatrix} -1.05 \\ 0.67 \\ 1.88 \\ -2.40 \end{bmatrix} + \begin{bmatrix} 0.36 \\ -0.65 \\ 0.36 \\ 1.5 \end{bmatrix} = \begin{bmatrix} -0.69 \\ 0.02 \\ 2.24 \\ -0.74 \end{bmatrix}$$

$$\tanh(w_c * I + B_c) = \tilde{c}_t = \begin{bmatrix} -0.59 \\ 0.01 \\ 0.97 \\ -0.62 \end{bmatrix}$$

$$f_t \circledast c_{t-1} = \begin{bmatrix} 0.93 \\ 0.26 \\ 0.07 \\ 0.57 \end{bmatrix} \circledast \begin{bmatrix} 0.69 \\ -0.18 \\ 0.66 \\ 1.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.64 \\ -0.64 \\ 0.04 \\ 0.74 \end{bmatrix}$$

$$i_t \circledast \tilde{c}_t = \begin{bmatrix} 0.33 \\ 0.50 \\ 0.90 \\ 0.32 \end{bmatrix} \circledast \begin{bmatrix} -0.59 \\ 0.01 \\ 0.97 \\ -0.62 \end{bmatrix}$$

$$= \begin{bmatrix} -0.19 \\ 0.005 \\ 0.873 \\ -0.1984 \end{bmatrix}$$

$$c_{t-1} = f_t \circledast c_{t-1} \oplus i_t \circledast \tilde{c}_t$$

$$\begin{bmatrix} 0.64 \\ -0.04 \\ 0.04 \\ 0.74 \end{bmatrix} \oplus \begin{bmatrix} -0.19 \\ 0.005 \\ 0.873 \\ -0.1984 \end{bmatrix} = \begin{bmatrix} 0.45 \\ -0.03 \\ 0.91 \\ 0.54 \end{bmatrix}$$

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$$O_t = \text{Sigmoid}(W_o * I + B_o)$$

$W_o * I$

$$= \begin{bmatrix} 0.26 & -0.074 & -1.9 & -0.027 & 0.06 & 2.5 & -0.19 & 0.3 \\ -0.035 & -1.2 & 1.1 & 0.75 & 0.79 & -0.91 & 1.4 & -1.4 \\ 0.59 & 2.2 & -0.99 & -0.57 & 0.1 & -0.5 & -1.6 & 0.06 \\ -1.1 & 0.47 & -0.92 & 1.5 & -0.78 & -0.32 & 0.81 & -1.2 \end{bmatrix}$$

$$* \begin{bmatrix} 0.51 \\ -0.12 \\ 0.03 \\ 0.16 \\ -1.38 \\ -0.94 \\ 0.52 \\ 0.51 \end{bmatrix} = \begin{bmatrix} -2.29 \\ 0.05 \\ -0.54 \\ 0.78 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + B_o)$$

$$W_o * I + B_o = \begin{bmatrix} -2.29 \\ 0.05 \\ -0.54 \\ 0.78 \end{bmatrix} + \begin{bmatrix} 0.23 \\ 1.3 \\ -1.6 \\ 0.18 \end{bmatrix}$$

$$\begin{bmatrix} -2.06 \\ 1.35 \\ -2.14 \\ 0.96 \end{bmatrix}$$

$$O_t = \text{Sigmoid}(W_o * I + B_o) = \begin{bmatrix} 0.11 \\ 0.79 \\ 0.10 \\ 0.72 \end{bmatrix}$$

$$C_{t-1} = \tanh\begin{pmatrix} 0.45 \\ -0.03 \\ 0.91 \\ 0.54 \end{pmatrix} = \begin{pmatrix} 0.42 \\ -0.02 \\ 0.72 \\ 0.49 \end{pmatrix}$$

$$h_t = O_t * C_t$$

$$h_t = \begin{bmatrix} 0.11 \\ 0.79 \\ 0.10 \\ 0.72 \end{bmatrix} * \begin{bmatrix} 0.42 \\ -0.02 \\ 0.72 \\ 0.49 \end{bmatrix}$$

$$= \begin{bmatrix} 0.04 \\ -0.01 \\ 0.07 \\ 0.35 \end{bmatrix}$$

$$h_{t-1} = \begin{bmatrix} 0.04 \\ -0.01 \\ 0.07 \\ 0.35 \end{bmatrix}$$

$$C_{t-1} = \begin{bmatrix} 0.45 \\ -0.03 \\ 0.91 \\ 0.54 \end{bmatrix}$$

softmax

Output layer = [W(output layer) * h_{t-1} + bias]

$$\begin{bmatrix} 0.26 & 0.78 & -1.2 & -1.3 \\ 0.52 & 0.3 & 0.25 & 0.35 \\ -0.68 & 0.23 & 0.29 & -0.71 \\ 1.9 & 0.47 & -1.2 & 0.66 \\ -0.97 & 0.79 & 1.2 & -0.82 \\ 0.96 & 0.41 & 0.82 & 1.9 \\ -0.25 & -0.75 & -0.89 & -0.82 \\ -0.077 & 0.34 & 0.28 & 0.83 \end{bmatrix} * \begin{bmatrix} 0.04 \\ -0.01 \\ 0.07 \\ 0.35 \end{bmatrix}$$

$$\begin{bmatrix} -0.53 \\ 0.15 \\ -0.25 \\ 0.21 \\ -0.24 \\ 0.75 \\ -0.35 \\ 0.30 \end{bmatrix} + \begin{bmatrix} 0.013 \\ 1.5 \\ -0.26 \\ 2.7 \\ 0.63 \\ -0.86 \\ -1.1 \\ 0.48 \end{bmatrix}$$

$$= \begin{bmatrix} -0.517 \\ 1.65 \\ -0.51 \\ 2.91 \\ 0.39 \\ -0.11 \\ -1.45 \\ 0.78 \end{bmatrix} = \text{Softmax}\begin{bmatrix} -0.517 \\ 1.65 \\ -0.51 \\ 2.91 \\ 0.39 \\ -0.11 \\ -1.45 \\ 0.78 \end{bmatrix}$$

$e^{-0.517} = 0.59$, $e^{1.65} = 5.20$, $e^{-0.51} = 0.60$

$e^{2.91} = 18.35$, $e^{0.39} = 1.47$, $e^{-0.11} = 0.89$

$e^{-1.45} = 0.23$, $e^{0.78} = 2.18$

$\sum e^{z_j} = 0.59 + 5.20 + 0.60 + 18.35 + 1.47 + 0.89 + 0.23 + 2.18$

CamScanner

0.59 / 29.51 = 0.01
5.20 / 29.51 = 0.17
0.60 / 29.51 = 0.02
18.35 / 29.51 = 0.62
1.47 / 29.51 = 0.04
0.89 / 29.51 = 0.03
0.23 / 29.51 = 0.007
2.18 / 29.51 = 0.07

0.62 is the highest Probability on fourth output neuron.

So the next word is "chair"

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	Highest prob	Predicted word
Average embedding	0.60	chair
Target embedding	0.54	chair
Context embedding	0.65	chair
LSTM forward pass with all given words	0.62	chair

All these (three) embedding give identical result.

Moreover I also check LSTM forward pass with all given words its result is also identical with other three embeddings.

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